

AI & ML Internship: Task 5 Report

Project: Ensemble Learning, Model Optimization & Real-World ML Pipeline

1. Objective

The objective of this task was to build a complete machine learning classification pipeline using the Breast Cancer Dataset. To mirror real-world ML systems where single models are rarely enough, this project transitioned from a baseline model to advanced Ensemble Learning techniques (Random Forest and Gradient Boosting). Finally, the system's production performance was improved through systematic Hyperparameter Tuning and Cross-Validation.

2. Model Optimization Strategy

In production environments, optimization is critical because it significantly improves a model's performance and stability.

To optimize the Random Forest model, **GridSearchCV** was utilized to perform automated hyperparameter tuning.

- Instead of relying on default settings, the grid search systematically tested multiple combinations of the `n_estimators` (50, 100) and `max_depth` (3, 5, 7) parameters.
- To ensure reliable evaluation and prevent the model from simply memorizing a single train-test split, the tuning process was paired with **3-Fold Cross-Validation**.
- **Optimal Parameters:** The grid search identified that the best hyperparameters for this specific dataset were [Insert best `n_estimators` here] estimators with a maximum depth of [Insert best `max_depth` here].

3. Model Comparison: Baseline vs. Ensemble Models

Three models were trained and evaluated using Accuracy, Precision, Recall, and F1-score to analyze their comparative strengths.

- **Logistic Regression (Baseline):** The baseline linear model performed reasonably well but inherently struggles to capture highly complex, non-linear biological relationships present in medical data.
- **Random Forest Classifier:** This is a powerful ensemble method that utilizes "bagging" (combining multiple decision trees) to inherently reduce overfitting. By averaging the predictions of many independent trees, it creates a highly stable classification boundary.

- **Gradient Boosting Classifier:** This advanced ensemble technique utilizes "boosting," which builds strong models from weak learners by sequentially correcting the errors of the previous trees.

Metrics Summary:

- **Logistic Regression:** Accuracy = [Value], Precision = [Value], Recall = [Value], F1-Score = [Value]
- **Random Forest:** Accuracy = [Value], Precision = [Value], Recall = [Value], F1-Score = [Value]
- **Gradient Boosting:** Accuracy = [Value], Precision = [Value], Recall = [Value], F1-Score = [Value]

4. Final Model Selection

The final model was selected based on a strict combination of three factors: **Performance, Stability, and Interpretability.**

The **Tuned Random Forest Classifier** was selected as the final optimized model.

- **Performance & Stability:** Ensemble models inherently provide higher accuracy and stability than single models. By combining the Random Forest architecture with GridSearchCV, we successfully constrained the tree depths, reducing variance and ensuring the model will generalize exceptionally well to unseen patient data.
- **Interpretability (Feature Importance):** In healthcare predictions, interpretability is a key requirement. The Random Forest model allowed us to extract Feature Importances, revealing exactly which cellular measurements (e.g., radius, perimeter, concavity) the algorithm relied on most heavily to detect malignancy.